

Hodge Theory on $T^4/\mathbb{Z}_3$ within FFGFT

Frobenius, Sector Pairing and Algebraic Cycles —
a Known Special Case and Its Galois Structure

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Hodge Theory on T^4/Z_3 within FFGFT

Abstract

The second Millennium Problem of mathematics — the Hodge Conjecture — asks whether every Hodge class of a smooth projective variety is a rational linear combination of algebraic cycle classes. For the base geometry of FFGFT, the orbifold $T^4/\mathbb{Z}_3$, this question admits a positive answer: the complex 2-torus T^4 is polarisable in the form $E_\omega \times E_\omega$ and hence an abelian surface for which the Hodge Conjecture is known to hold; the nine $\mathbb{Z}_3$ fixed points furnish algebraic 0-cycles; and the sector pairing $k \leftrightarrow -k$ (Doc. 343) is the Hodge symmetry $H^{p,q} \leftrightarrow H^{q,p}$. The Galois Frobenius $x \mapsto x^3$ on $GF(27)^*$, by contrast, is not identical to the algebraic Frobenius on étale cohomology: the latter's eigenvalues on H^1 have absolute value $\sqrt{3}$ and are not roots of unity. The Millennium Problem in its general form — for arbitrary smooth projective varieties — remains open. What FFGFT supplies is structural completeness for its own geometry.

Status Markers

[B]	Algebraically proved.
[K]	Numerically confirmed.
[S]	Structurally plausible; physical identification open.
[E]	Established external mathematics.
[Q]	Taken from the literature, correctly cited.
[SETZUNG]	Axiomatic stipulation.

1 The Millennium Problem

The Clay Mathematics Institute states the Hodge Conjecture as follows **[Q]**: On a non-singular projective algebraic variety X over $\mathbb{C}$, any Hodge class — that is, any cohomology class of type (k, k) — is a rational linear combination of classes of algebraic cycles of codimension k .

More precisely: the Hodge decomposition

$$H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X)$$

splits cohomology by type; Hodge classes are the elements of $H^{k,k}(X) \cap H^{2k}(X, \mathbb{Q})$. The conjecture states that these classes are generated by genuine geometric objects — subvarieties whose cohomology classes are rationally combined. For smooth projective curves and surfaces in bidegree $(1, 1)$ this follows from the Lefschetz theorem on $(1, 1)$ -classes **[E]**. In general dimension the conjecture is open.

2 The Complex 2-Torus as an Abelian Surface

The real geometry $T^4 = S^1 \times S^1 \times S^1 \times S^1$ carries in FFGFT the structure of a *complex 2-torus*: two complex coordinate pairs $(z_1, z_2) \in \mathbb{C}^2/\Lambda$, where Λ is the mode lattice **[SETZUNG]** (Doc. 001).

A polarisable complex 2-torus — here $E_\omega \times E_\omega$ — is an abelian surface, an abelian variety of complex dimension 2. By the Riemann conditions $\mathbb{C}^2/\Lambda$ is projective only with a polarisation; the generic 2-torus is not algebraic, whereas $E_\omega \times E_\omega$ as a product of elliptic curves is. For abelian varieties the Hodge Conjecture is known in the cases relevant here **[E]** **[Q]**:

- For abelian surfaces (complex dimension 2) the conjecture holds completely for all Hodge classes: $H^{1,1}$ is covered by the Lefschetz theorem, $H^{2,2}$ is generated by the point class.
- Products of elliptic curves and tori over $\mathbb{Q}$ are fully spanned by algebraic cycles.

[B] *Theorem A*: T^4 as the abelian surface $E_\omega \times E_\omega$ satisfies the Hodge Conjecture: all (k, k) -classes are rational linear combinations of algebraic cycles.

Proof: In complex dimension 2, (k, k) -classes exist only for $k = 0, 1, 2$ (Hodge diamond $h^{p,q} = \binom{2}{p}\binom{2}{q}$, $h^{1,1} = 4$). For $k = 0$ and $k = 2$: trivial via the fundamental class ($H^{0,0}$) and the point class ($H^{2,2}$), respectively. For $k = 1$: Lefschetz theorem on $(1, 1)$ -classes (known **[E]**). $H^{2,2}(T^4)$ is moreover generated by products of $(1, 1)$ -classes, hence algebraic. $\square$

3 The Nine $\mathbb{Z}_3$ Fixed Points as Algebraic 0-Cycles

Under the $\mathbb{Z}_3$ action on T^4 there are exactly nine fixed points (Doc. 001/330) **[B]**. These fixed points are algebraic 0-cycles — points on the variety $T^4/\mathbb{Z}_3$ with well-defined cohomology classes.

[B] Theorem B: The nine $\mathbb{Z}_3$ fixed points of $T^4/\mathbb{Z}_3$ are algebraic 0-cycles of codimension 4. Their cohomology classes generate the twisted sector of the orbifold cohomology. In the orbifold cohomology (Chen–Ruan) **[Q]**:

$$H_{\text{CR}}^*(T^4/\mathbb{Z}_3) = H^*(T^4)^{\mathbb{Z}_3} \oplus \bigoplus_{9 \text{ fixed points}} H^*(\{p\})$$

The second summand is algebraic: every point is an algebraic 0-cycle; its cohomology class is trivially algebraic. **[E]**

Under the Witt-pair structure of Doc. 341:

- **Even indices 0,2,4,6** = $\mathbb{Z}_3$ fixed points = algebraic 0-cycles **[B]**
- **Odd indices 1,3,5,7,8** = three-cycles = higher-dimensional algebraic cycles **[B]**

This is literally the Hodge statement: topological class = algebraic cycle, for all classes in the twisted sector.

4 Frobenius Automorphism and Algebraic Frobenius

This section separates the Galois Frobenius on $GF(27)^*$ from the algebraic Frobenius of étale cohomology.

The Galois Frobenius on $GF(27)^*$

Doc. 339 proves **[B]**: the Frobenius automorphism on $GF(27) = GF(3^3)$ is $x \mapsto x^3$. It acts on $GF(27)^* \cong \mathbb{Z}_{26}$ and generates the sector structure: fixed points $\{+1, -1\}$ (massive sector), eight three-cycles (gluons/massless sector).

The Algebraic Frobenius on Étale Cohomology

In the arithmetic geometry of algebraic geometry **[Q]**, for a variety X over $GF(q)$ the geometric Frobenius is $F_q : x \mapsto x^q$ on points. It induces an endomorphism F_q^* on the ℓ -adic cohomology $H_{\text{ét}}^i(X, \mathbb{Q}_\ell)$. The eigenvalues of F_q^* are algebraic numbers that, by the Weil–Deligne theorem **[E]**, determine the zeta function and the Betti numbers of X (weights: $|\alpha| = q^{i/2}$ on H^i). The Hodge numbers $h^{p,q}$ are not determined by them.

Theorem C (no identification): The Galois Frobenius $x \mapsto x^3$ on $GF(27)^*$ is not identical to the algebraic Frobenius on $H_{\text{ét}}^1$ of $T^4/\mathbb{Z}_3$ over $GF(3)$. The obvious identification fails because, by Weil, the Frobenius eigenvalues on H^1 of an abelian surface over $\mathbb{F}_q$ have absolute value $\sqrt{q}$, here $\sqrt{3} \approx 1.732$: they are four Weil numbers, not roots of unity and not 26 of them. Example: $y^2 = x^3 - x$ over $GF(3)$ has $\#E = 4$ and eigenvalues $\pm i\sqrt{3}$ (cross-check $\#E(GF(27)) = 28$); for $E \times E$ there are four eigenvalues with $\alpha^{26} = -3^{13}$. An identification with the elements of $GF(27)^*$ does not follow. The two share only the map $x \mapsto x^3$ as generator of $\text{Gal}(GF(27)/GF(3))$.

[S] Remark: The physical interpretation — Frobenius orbits as particle sectors — is an FFGFT classification and not part of the mathematical proof.

5 Sector Pairing as Hodge Symmetry

Doc. 343, Theorem D **[B]** proves: the sector pairing $k \leftrightarrow -k \pmod{13}$ on the primitive $GF(27)^*$ classes pairs $f_1 \leftrightarrow f_2$ and $f_3 \leftrightarrow f_4$.

[B] Theorem D: This sector pairing is the algebraic realisation of the Hodge symmetry $H^{p,q} \leftrightarrow H^{q,p}$ under complex conjugation.

Proof: The Hodge decomposition on T^4 has $H^{1,0}$ and $H^{0,1}$ as a conjugate pair, and $H^{2,1}$ and $H^{1,2}$ likewise. Complex conjugation $\bar{\cdot} : H^{p,q} \rightarrow H^{q,p}$ is the algebraic datum selecting real cohomology classes. On $GF(27)^*$ the corresponding map is inversion $g^k \mapsto g^{-k}$, i.e. negation of the exponent in $\mathbb{Z}_{26}$. It must be distinguished from field negation $e \mapsto -e = e \cdot g^{13}$: that map sends every order-26 orbit to an order-13 orbit and is therefore not a pairing within the primitive classes (check script pruef_363b, part L3). Inversion, by contrast, is an involution on the four primitive classes with no fixed orbit; it reduces $\{f_1, f_2, f_3, f_4\}$ to two real classes — exactly the Hodge symmetry. $\square$

From Doc. 343 Theorem D''' **[B]**: $\{f_3, f_4\}$ is the Dirac layer (charged leptons), $\{f_1, f_2\}$ the Majorana layer (neutrinos). The Hodge pairing separates exactly the physical sectors.

6 The χ -Classes as Algebraic Basis

Doc. 342 **[B]** classifies the eight irreducible cubic polynomials over $GF(3)$:

- Four primitive polynomials of order 26: f_1, f_2, f_3, f_4
- Four polynomials of order 13: g_1, g_2, g_3, g_4

[B] Theorem E: The eight polynomials generate the factorisation classes of $GF(3)[x]/(x^{26}-1)$ completely (Doc. 342).

A reading of these classes as a $\mathbb{Q}$ -basis of algebraic cycles on $T^4/\mathbb{Z}_3$ is not established thereby: it would require the irreducible factors of the characteristic polynomial of F_3^* on H^i to be the eight polynomials of Doc. 342, i.e. the identification of Theorem C, which does not hold.

7 The Hodge Conjecture for $T^4/\mathbb{Z}_3$

[B] Main Theorem: All Hodge classes on $T^4/\mathbb{Z}_3$ are rational linear combinations of algebraic cycles.

Proof from Theorems A and B:

1. $H^*(T^4)^{\mathbb{Z}_3}$ (untwisted sector): All Hodge classes come from the $\mathbb{Z}_3$ -invariant part of $H^*(T^4)$. Since T^4 as an abelian surface satisfies the Hodge Conjecture (Theorem A) and the $\mathbb{Z}_3$ projection maps algebraic cycles to algebraic cycles, all invariant Hodge classes are algebraic.
2. Twisted sector: The nine fixed points are algebraic 0-cycles (Theorem B); their cohomology classes are trivially algebraic.
3. By the Chen–Ruan decomposition (Section 3), orbifold cohomology consists of exactly these two summands; no Hodge class lies outside them. The Frobenius structure is not needed for this. $\square$

In plain language: For the specific geometry of FFGFT the Hodge Conjecture is not an open problem. It follows from combining the abelian-surface theorem and orbifold cohomology. The Millennium Problem in its general form is not solved by this.

8 What FFGFT Does Not Supply

General varieties. The Hodge Conjecture is open for smooth projective varieties of arbitrary dimension. $T^4/\mathbb{Z}_3$ is a very special class — a torus orbifold with rich algebraic structure. For a general variety of Hodge type (2, 2) in dimension 4 none of the above arguments apply.

No singularities in the FFGFT reading. FFGFT does not work on the quotient space $T^4/\mathbb{Z}_3$ as a point set, but on the smooth T^4 with $\mathbb{Z}_3$ as a lattice symmetry of the modes (Doc. 314: geometry $\rightarrow$ spectrum + fibre; the orbifold is a lattice symmetry, not an additional assumption). The cohomology is the $\mathbb{Z}_3$ -equivariant cohomology of a smooth abelian surface $E_\omega \times E_\omega$; Theorem A applies directly, with no detour through singularities. **[B]**

If one nevertheless forms the quotient space, the $\mathbb{Z}_3$ action has, by Doc. 330, eigenvalues $\{\omega, \omega^2\}$ on $\mathbb{C}^2$, hence $\det = 1$: it lies in $SU(2)$. The nine fixed points are then A_2 points and admit a crepant resolution; the result is a K3 surface — smooth, projective, and for K3 surfaces the Hodge Conjecture is known (Lefschetz on $H^{1,1}$ suffices in dimension 2) **[E]**. In neither reading is there an obstruction.

Premise and level of representation. Higher-dimensional spaces and objects may be constructed freely as long as they are translations of the T^4 content (Type III, Doc. 270/330 **[K]**): they contain no more than T^4 , and no projection from them can yield more than the mode content of T^4 (Doc. 314: lattice = index set of Fourier modes; Doc. 343: the resolution limit lies in the object itself). All theorems of this document are such translations — the same compact geometry in the Galois language of $GF(27)^*$ and in the language of the Hodge decomposition, bijective and lossless. The document contains no Type I or Type II operation (duality decompactification, geometric projection) and therefore makes no claim about the observability of individual cycles; that is governed by the projection theory (Doc. 285, 330, 333).

Non-commutative extension. Orbifold cohomology (Chen–Ruan) agrees with the classical Hodge decomposition in the untwisted sector; the twisted sector has a grading shift modifying the Hodge symmetry. The statement here holds for the classical picture; the Chen–Ruan extension is to be rated **[S]**.

9 Summary

FFGFT structure	Hodge dence	correspon-	Doc.	Status
T^4 as abelian 2-torus	Hodge known	conjecture	001	[E]
9 $\mathbb{Z}_3$ fixed points	Algebraic 0-cycles		001/330	[B]
Frobenius $x \mapsto x^3$ on $GF(27)^*$	No identification with the Frobenius on H_{et}^1 (eigenvalues of absolute value $\sqrt{3}$)		339/341	—
Sector pairing $k \leftrightarrow -k$	Hodge $H^{p,q} \leftrightarrow H^{q,p}$	symmetry	343	[B]
χ -classes (8 polynomials)	Factorisation classes of $GF(3)[x]/(x^{26}-1)$; no cycle basis		342	[B]
Hodge conjecture for $T^4/\mathbb{Z}_3$	All Hodge classes algebraic		—	[B]
General Hodge conjecture	Millennium Problem open		—	—

The Hodge Conjecture is not an open problem for $T^4/\mathbb{Z}_3$. It follows from Theorem A and orbifold cohomology. Of the Galois structure of $GF(27)^*$, the sector pairing realises the Hodge symmetry (Theorem D); the Galois Frobenius and the χ -classes, by contrast, cannot be identified with the Frobenius or the cycles of étale cohomology (Theorem C). The general Millennium Problem is unaffected.